

WhatsApp POS Receipt Integration Prompt - 2026-08-01

Act as a multidisciplinary principal implementation team: system architect, cyber-security architect, senior backend engineer, senior frontend engineer, UI/UX specialist, QA lead, compliance reviewer, and SaaS growth advisor.

Implement a fresh, enterprise-grade WhatsApp receipt integration for Stoquify's POS sales/payment workflow. Treat this as a controlled new integration, not a casual rollback reversal. The integration must be professional, secure, scalable, auditable, and designed so future WhatsApp features can be added later without rewriting the POS receipt system.

Goal

Allow POS cashiers to send sales/payment receipts through WhatsApp after a completed POS sale, while preserving the existing PRINT, EMAIL, SMS, and NONE receipt behavior. WhatsApp must be a delivery channel only, never a source of truth.

Primary Scope

- POS sale completion flow
- POS payment receipt delivery
- Receipt channel schema and service contract
- WhatsApp receipt provider abstraction
- WhatsApp Business Cloud API adapter or safe provider boundary
- POS UI channel selection and consent state
- Receipt delivery audit trail
- Outbox/queue-safe delivery behavior
- Environment/config validation
- Focused tests and evidence report

Design Requirements

- Keep sale completion, payment capture, stock movement, accounting posting, fiscalization, and audit as the source of truth.
- Send WhatsApp receipts only after the POS sale has been safely committed.
- Do not let WhatsApp delivery failure roll back a completed sale.
- Require customer phone number or explicit cashier-entered destination.
- Require explicit customer consent before WhatsApp delivery.
- Use approved template-based messaging where live WhatsApp sends are enabled.
- Keep live sends disabled by default unless all required config and flags are present.
- Support stub/sandbox behavior for local development and tests.
- Store delivery attempt evidence with channel, status, destination hash or redacted destination, provider reference, retryability, and safe error message.
- Do not log access tokens, full phone numbers, message bodies with sensitive data, or secrets.

Security And Compliance Rules

- No hard-coded WhatsApp credentials.
- No raw secrets in reports, logs, tests, screenshots, or audit rows.
- Validate and normalize phone numbers before delivery.
- Redact phone numbers in UI where appropriate.
- Enforce RBAC, tenant isolation, module entitlement, fresh auth where required, and audit boundaries.
- WhatsApp must not bypass fiscal receipt legal-delivery rules.
- If receipt delivery is legally blocked until fiscal certification, WhatsApp must also be blocked.
- Preserve existing non-WhatsApp receipt behavior.
- Do not weaken existing policy gates.

Architecture Requirements

- Introduce or extend a provider-neutral receipt delivery interface.
- Add WhatsApp as a channel through a clean adapter boundary.
- Separate receipt payload generation, delivery eligibility and consent validation, provider-specific WhatsApp send logic, audit/evidence recording, and UI channel selection.
- Make the design extensible for future WhatsApp workflows such as payment reminders, customer statements, supplier notifications, approval nudges, close evidence requests, and support messages.
- Avoid overbuilding future features now; only create the minimal clean foundation needed for receipts.

Execution Rules

- Inspect current POS receipt code before editing.
- Read the latest rollback/refinement reports under `what - next /`.
- Do not use destructive git commands.
- Do not revert unrelated user changes.
- Keep changes narrow and traceable to POS WhatsApp receipt delivery.
- If the current baseline conflicts with the requested integration, stop and produce a blocker report instead of guessing.

Implementation Checklist

- 1 Map the current POS receipt workflow end to end.
- 2 Define the WhatsApp delivery contract and provider boundary.
- 3 Add `WHATSAPP` to receipt channel handling in a controlled way.
- 4 Add consent and destination validation.
- 5 Add safe environment/config parsing for WhatsApp provider settings.
- 6 Implement stub/sandbox provider behavior first.
- 7 Implement live-provider boundary only if credentials/config gates are safe.
- 8 Update POS UI with a professional WhatsApp receipt option and consent requirement.
- 9 Ensure delivery failure is non-blocking after sale commit.

- 10 Record delivery audit/evidence safely.
- 11 Add focused tests for WhatsApp schema, missing destination, missing consent, legal delivery blocking, sale commit non-rollback, no live send without config, UI gating, and no secret/raw phone leakage.
- 12 Run focused POS/receipt tests.
- 13 Run relevant policy/module gates if touched.
- 14 Save an implementation evidence report under
`what-next/WHATSAPP_POS_RECEIPT_INTEGRATION_REPORT_YYYY-MM-DD.md`.

Success Criteria

- POS supports WhatsApp receipt delivery as a controlled channel.
- Existing PRINT, EMAIL, SMS, and NONE behavior still works.
- WhatsApp sends require valid destination and consent.
- Live sending is disabled unless safely configured.
- Delivery failures do not roll back completed sales.
- Audit/evidence is recorded without leaking secrets or raw sensitive data.
- Focused tests pass.
- The foundation can support future WhatsApp features without rewriting POS receipt delivery.